

Qualification Pack



Biomass Pellet Manufacturing Junior Technician

QP Code: SGJ/Q4201

Version: 1.0

NSQF Level: 3

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Qualification Pack

Contents

SGJ/Q4201: Biomass Pellet Manufacturing Junior Technician	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SGJ/N4013: Basics of Biomass availability and utilization	5
SGJ/N4014: Preparation of Raw material and quality assessment	9
SGJ/N4015: Equipment and machineries of Biomass pellet plant	13
SGJ/N4016: Manufacturing of Biomass pellet	17
SGJ/N4017: Operation and maintenance of various components of biomass pellet plant	21
SGJ/N4018: Package and Storage of biomass pellet	25
SGJ/N4051: Maintain Personal Health & Safety in biomass pellet manufacturing facility	29
DGT/VSQ/N0101: Employability Skills (30 Hours)	33
Assessment Guidelines and Weightage	38
<i>Assessment Guidelines</i>	38
<i>Assessment Weightage</i>	39
Acronyms	40
Glossary	41

Qualification Pack

SGJ/Q4201: Biomass Pellet Manufacturing Junior Technician

Brief Job Description

Biomass Pellet Manufacturing Junior Technician is responsible for production, storage, and management of biomass pellets at the plant. He/she looks after activities such as biomass receiving, pre-processing/densification and storage in a manner to maintain its quality and safety as per standards.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it without any accidents so diligence and hardworking are desired attributes for individuals performing this role. He/She must also demonstrate strong work ethics, an ability to communicate courteously with co-workers and must be good with following instructions of the supervisor

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4013: Basics of Biomass availability and utilization](#)
2. [SGJ/N4014: Preparation of Raw material and quality assessment](#)
3. [SGJ/N4015: Equipment and machineries of Biomass pellet plant](#)
4. [SGJ/N4016: Manufacturing of Biomass pellet](#)
5. [SGJ/N4017: Operation and maintenance of various components of biomass pellet plant](#)
6. [SGJ/N4018: Package and Storage of biomass pellet](#)
7. [SGJ/N4051: Maintain Personal Health & Safety in biomass pellet manufacturing facility](#)
8. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
Country	India

Qualification Pack

NSQF Level	3
Credits	10
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8142.1301 Plastic Moulding Technician or Operator
Minimum Educational Qualification & Experience	10th grade pass OR 8th grade pass (with two year of (NTC/ NAC) after 8th) with NA of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 2.5) with 1-2 Years of experience 1.5-year relevant experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-03-ES-00911-2023-V1-SCGJ
NQR Version	1.0

Remarks:

Total 300 Hours i.e., 10 Credits (Including 210 Hours of domain NOS+ 30 hours of Employability+ 60 hours of On the Job Training OJT)

Qualification Pack

SGJ/N4013: Basics of Biomass availability and utilization

Description

This unit outlines the basics of biomass and how the resource is to be utilised for upstream process

Scope

The scope covers the following :

- Introduction to biomass sector Opportunities
- understand the importance of agriculture waste as renewable source of energy

Elements and Performance Criteria

Basics of Biomass availability and utilization

To be competent, the user/individual on the job must be able to:

- PC1.** Provide introduction to the domestic Biomass sector
- PC2.** Provide an overview of biomass potential in India
- PC3.** Describe different types of farm waste and its source, supply chain and storage methodology. and show different types of farm waste and its source, supply chain and storage methodology through pictures and videos.
- PC4.** Explain the importance of agriculture residue waste as a sustainable energy resource and its management and end use applications
- PC5.** Explain about biomass supply chain and its challenges
- PC6.** Discuss biomass energy conversion technologies and role of biomass pellets as green fuel and illustrate biomass energy conversion technologies and role of biomass pellets as green fuel.
- PC7.** Explain the terminologies used in pellet manufacturing plant sites
- PC8.** identify the role and responsibilities of Biomass Pellet Manufacturing Junior Technician.
- PC9.** discuss advantage of doing this course, future prospect, and career opportunities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** benefits to the company due to practice of these procedures
- KU5.** the importance of having an ethical and value based approach to governance
- KU6.** knowledge of processes involved in the business
- KU7.** country specific regulations for manufacturing and its importance
- KU8.** reporting procedure in case of deviations
- KU9.** different types of biomass available

Qualification Pack

KU10. applications of biomass

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** follow organization's rule-based decision making process
- GS8.** follow organisation's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Biomass availability and utilization</i>	40	10	-	-
PC1. Provide introduction to the domestic Biomass sector	2	-	-	-
PC2. Provide an overview of biomass potential in India	3	-	-	-
PC3. Describe different types of farm waste and its source, supply chain and storage methodology. and show different types of farm waste and its source, supply chain and storage methodology through pictures and videos.	5	5	-	-
PC4. Explain the importance of agriculture residue waste as a sustainable energy resource and its management and end use applications	5	-	-	-
PC5. Explain about biomass supply chain and its challenges	5	-	-	-
PC6. Discuss biomass energy conversion technologies and role of biomass pellets as green fuel and illustrate biomass energy conversion technologies and role of biomass pellets as green fuel.	5	5	-	-
PC7. Explain the terminologies used in pellet manufacturing plant sites	5	-	-	-
PC8. identify the role and responsibilities of Biomass Pellet Manufacturing Junior Technician.	5	-	-	-
PC9. discuss advantage of doing this course, future prospect, and career opportunities	5	-	-	-
NOS Total	40	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4013
NOS Name	Basics of Biomass availability and utilization
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4014: Preparation of Raw material and quality assessment

Description

This unit outlines how to prepare raw material and perform quality assessment for biomass pellet production

Scope

The scope covers the following :

- various features of raw material in use for biomass pellet production
- different parameters of feedstock for biomass pellet manufacturing

Elements and Performance Criteria

Preparation of Raw material and quality assessment

To be competent, the user/individual on the job must be able to:

- PC1.** List the raw materials to be used and show samples of different types of raw materials and binders in use
- PC2.** Describe the process of receiving, handling and storage of raw materials and demonstrate the storage procedures for raw materials such as saw dust, wheat straw, etc
- PC3.** Assess different parameters of feedstock for biomass pellet
- PC4.** Explain how to pre-process feedstock and prepare the raw material for feeding
- PC5.** Demonstrate how to measure moisture content of raw material
- PC6.** Show how to check the raw material quality and grade
- PC7.** show how to calculate batch size and prioritize urgent orders based on the production schedule and machine capacity
- PC8.** Show how to inspect the conformance of raw material quality to standards

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures that impact the business process
- KU2.** legal formalities applicable for the particular business
- KU3.** benefits to the company due to practice of these procedures
- KU4.** the importance of having an ethical and value based approach to governance
- KU5.** knowledge of processes involved in the business
- KU6.** country specific regulations for manufacturing and its importance
- KU7.** reporting procedure in case of deviations
- KU8.** knowledge of types of biomass
- KU9.** applications of biomass

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organisation's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparation of Raw material and quality assessment</i>	23	27	-	-
PC1. List the raw materials to be used and show samples of different types of raw materials and binders in use	5	4	-	-
PC2. Describe the process of receiving, handling and storage of raw materials and demonstrate the storage procedures for raw materials such as saw dust, wheat straw, etc	5	4	-	-
PC3. Assess different parameters of feedstock for biomass pellet	5	-	-	-
PC4. Explain how to pre-process feedstock and prepare the raw material for feeding	3	3	-	-
PC5. Demonstrate how to measure moisture content of raw material	-	3	-	-
PC6. Show how to check the raw material quality and grade	5	3	-	-
PC7. show how to calculate batch size and prioritize urgent orders based on the production schedule and machine capacity	-	5	-	-
PC8. Show how to inspect the conformance of raw material quality to standards	-	5	-	-
NOS Total	23	27	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4014
NOS Name	Preparation of Raw material and quality assessment
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4015: Equipment and machineries of Biomass pellet plant

Description

This unit outlines specification and standard operating procedure of all equipment and machinery by giving practical training.

Scope

The scope covers the following :

- components of biomass pellet plant
- specification and standard operating procedure of all equipment/machinery by giving practical training

Elements and Performance Criteria

Equipment and machineries of Biomass pellet plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and show how to Identify different components of biomass pellet plant
- PC2.** discuss and show the working of infeed hopper, screw auger and baggage device
- PC3.** explain and show different types of dyes in use
- PC4.** explain and show the use of chipper, hammer mill and oven dryer and show them chipper, hammer mill and oven dryer.
- PC5.** discuss and List various tools, equipment and machinery used in biomass pellet plant and demonstrate how to operate equipment's/machine involved in manufacturing of pellets are dryer, hammer mill and pellet mill, etc
- PC6.** identify size, configuration and cost of key tools, machinery/equipment.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** benefits to the company due to practice of these procedures
- KU5.** the importance of having an ethical and value based approach to governance
- KU6.** knowledge of processes involved in the business
- KU7.** country specific regulations for manufacturing and its importance
- KU8.** reporting procedure in case of deviations
- KU9.** different types of biomass
- KU10.** applications and uses of biomass

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organisation's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Equipment and machineries of Biomass pellet plant</i>	30	20	-	-
PC1. discuss and show how to Identify different components of biomass pellet plant	5	-	-	-
PC2. discuss and show the working of infeed hopper, screw auger and baggage device	5	-	-	-
PC3. explain and show different types of dyes in use	5	-	-	-
PC4. explain and show the use of chipper, hammer mill and oven dryer and show them chipper, hammer mill and oven dryer.	5	10	-	-
PC5. discuss and List various tools, equipment and machinery used in biomass pellet plant and demonstrate how to operate equipment's/machine involved in manufacturing of pellets are dryer, hammer mill and pellet mill, etc	5	10	-	-
PC6. identify size, configuration and cost of key tools, machinery/equipment.	5	-	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4015
NOS Name	Equipment and machineries of Biomass pellet plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4016: Manufacturing of Biomass pellet

Description

This unit outlines all the process involved in manufacturing of biomass pellets

Scope

The scope covers the following :

- ALL Process involve in manufacturing of biomass pellets

Elements and Performance Criteria

Manufacturing of biomass pellets

To be competent, the user/individual on the job must be able to:

- PC1.** Explain about placing of biomass on dyes and demonstrate placing of biomass on dyes
- PC2.** Explain extrusion process
- PC3.** Explain about operation of various machines in plant such as dryer, hammer mill, pellet mill, bomb calorimeter, etc and demonstrate use of die, dryer, chipper, hammer mill, pellet mill, bomb calorimeter, etc
- PC4.** Show how to Set the control parameters during manufacturing
- PC5.** Explain about speed, temperature and moisture control in pelleting process and show how to Adjust pellet producing machineries at dye of 3mm, specific speed, temperature and moisture control
- PC6.** Explain the process of cooling of biomass pellet and discuss what is backpressure
- PC7.** Explain about standard protocol for testing of pellets and important parameters are Gross Calorific Value, Ash and Moisture.
- PC8.** Explain Do's and Don't
- PC9.** Demonstrate use of moisture meter to measure water content in different agriculture waste.
- PC10.** Perform the post-production cleaning and maintenance process followed in the industry.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** benefits to the company due to practice of these procedures
- KU5.** the importance of having an ethical and value based approach to governance
- KU6.** knowledge of processes involved in the business
- KU7.** country specific regulations for manufacturing and its importance
- KU8.** reporting procedure in case of deviations

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organisation,s code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manufacturing of biomass pellets</i>	30	20	-	-
PC1. Explain about placing of biomass on dyes and demonstrate placing of biomass on dyes	3	3	-	-
PC2. Explain extrusion process	3	-	-	-
PC3. Explain about operation of various machines in plant such as dryer, hammer mill, pellet mill, bomb calorimeter, etc and demonstrate use of die, dryer, chipper, hammer mill, pellet mill, bomb calorimeter, etc	3	3	-	-
PC4. Show how to Set the control parameters during manufacturing	-	3	-	-
PC5. Explain about speed, temperature and moisture control in pelleting process and show how to Adjust pellet producing machineries at dye of 3mm, specific speed, temperature and moisture control	3	3	-	-
PC6. Explain the process of cooling of biomass pellet and discuss what is backpressure	4	-	-	-
PC7. Explain about standard protocol for testing of pellets and important parameters are Gross Calorific Value, Ash and Moisture.	4	-	-	-
PC8. Explain Do's and Don't	3	-	-	-
PC9. Demonstrate use of moisture meter to measure water content in different agriculture waste.	3	4	-	-
PC10. Perform the post-production cleaning and maintenance process followed in the industry.	4	4	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4016
NOS Name	Manufacturing of Biomass pellet
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4017: Operation and maintenance of various components of biomass pellet plant

Description

This unit outlines the operation and maintenance activities in pellet plant.

Scope

The scope covers the following :

- Operation and Maintenance activities in pellet plant
- understand how to Maintain logbook for operation records and test results

Elements and Performance Criteria

Operation and maintenance of various components of biomass pellet plant

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss about the electricity in use for different stages of process
- PC2.** identify parameters to be mention in logbook and show how to Maintain logbook for operation records and test results.
- PC3.** explain how to ensure maintenance of all electrical and mechanical equipment's used in plant and demonstrate how to prepare routine maintenance and breakdown maintenance report
- PC4.** display the procedure to rectify faults and minor repairs in process machinery
- PC5.** explain "wash through" and "clogging" in pelleting process and demonstrate how to modify the moisture content or die dimensions to remove clogging of the process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** benefits to the company due to practice of these procedures
- KU5.** the importance of having an ethical and value based approach to governance
- KU6.** knowledge of processes involved in the business
- KU7.** country specific regulations for manufacturing and its importance
- KU8.** reporting procedure in case of deviations

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organisation's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operation and maintenance of various components of biomass pellet plant</i>	30	20	-	-
PC1. Discuss about the electricity in use for different stages of process	7	-	-	-
PC2. identify parameters to be mention in logbook and show how to Maintain logbook for operation records and test results.	5	10	-	-
PC3. explain how to ensure maintenance of all electrical and mechanical equipment's used in plant and demonstrate how to prepare routine maintenance and breakdown maintenance report	5	10	-	-
PC4. display the procedure to rectify faults and minor repairs in process machinery	5	-	-	-
PC5. explain "wash through" and "clogging" in pelleting process and demonstrate how to modify the moisture content or die dimensions to remove clogging of the process	8	-	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4017
NOS Name	Operation and maintenance of various components of biomass pellet plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4018: Package and Storage of biomass pellet

Description

This unit outlines key aspects of packaging and storage of biomass pellet

Scope

The scope covers the following :

- packaging and storage of biomass pellet
- labelling of the bags with types of biomass pellet and grade.

Elements and Performance Criteria

Package and Storage of biomass pellet

To be competent, the user/individual on the job must be able to:

- PC1.** discuss about the packaging and storage processes of finished product and show how to Carry out packing of finished product
- PC2.** discuss about of the temperature and moisture content of biomass pellet for packaging
- PC3.** discuss about the sizing of bags for packaging and show them the different sizes of bags available for packaging.
- PC4.** discuss labelling of the bags with types of biomass pellet and grade and show how to Label pellet bag with types of pellets and grade.
- PC5.** discuss about hazards associated with biomass storage and depict the hazards associated with storage of biomass through pictures and videos.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** benefits to the company due to practice of these procedures
- KU5.** the importance of having an ethical and value based approach to governance
- KU6.** knowledge of processes involved in the business
- KU7.** country specific regulations for manufacturing and its importance
- KU8.** reporting procedure in case of deviations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role

Qualification Pack

- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organisation's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Package and Storage of biomass pellet</i>	30	20	-	-
PC1. discuss about the packaging and storage processes of finished product and show how to Carry out packing of finished product	5	5	-	-
PC2. discuss about of the temperature and moisture content of biomass pellet for packaging	7	-	-	-
PC3. discuss about the sizing of bags for packaging and show them the different sizes of bags available for packaging.	6	5	-	-
PC4. discuss labelling of the bags with types of biomass pellet and grade and show how to Label pellet bag with types of pellets and grade.	6	5	-	-
PC5. discuss about hazards associated with biomass storage and depict the hazards associated with storage of biomass through pictures and videos.	6	5	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4018
NOS Name	Package and Storage of biomass pellet
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N4051: Maintain Personal Health & Safety in biomass pellet manufacturing facility

Description

This unit is about maintaining basic health and workplace safety in Biomass pellet manufacturing facility

Scope

The scope covers the following :

- Maintain standard health and workplace safety
- Organize safety awareness sessions and participate in safety drills
- Use and maintain protective safety gears

Elements and Performance Criteria

Maintain Personal Health & Safety in biomass pellet manufacturing facility

To be competent, the user/individual on the job must be able to:

- PC1.** identify the requirements for safe work area
- PC2.** explain how to Administer first aid and demonstrate how to Administer first aid.
- PC3.** . identify the personal protective equipment used for the specific purpose
- PC4.**
 - identify the heavy electrical and mechanical equipment's associated with manufacturing and
 - show how to Identify the heavy electrical and mechanical equipment's associated with
 - bioenergy production manufacturing
- PC5.**
 - identify work safety procedures and instructions to operate bioenergy production machines
 - and demonstrate how to Identify and perform work safety procedures and instructions to
 - operate bioenergy production machines
- PC6.**
 - explain occupational health & safety standards and regulations for bioenergy production
 - manufacturing process
- PC7.** demonstrate good housekeeping practices and infection control guidelines
- PC8.**
 - demonstrate how to Dispose- off any waste materials in accordance with safe working
 - practices and procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company occupational safety and health policy
- KU2.** company's medical policy
- KU3.** personal protective equipments (PPEs) required for safe operation
- KU4.** occupational health and safety (OHS) standards for workplace and machine safety
- KU5.** risk identification and mitigation procedures
- KU6.** workplace safety procedures including emergency evacuation, first aid procedures among others.

Qualification Pack

- KU7.** • understand work safety procedures and instructions to operate bioenergy production
• machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** how to be able to read safety instructions and warning signage
- GS2.** fill up documentation applicable to ones role
- GS3.** read instruction and safety manuals.
- GS4.** express statements or information clearly so that others can hear and understand
- GS5.** participate in and understand the main points of simple discussions
- GS6.** effectively communicate the danger.
- GS7.** make decisions pertaining to safety procedures at workplace
- GS8.** identify need of first aid and render it accordingly.
- GS9.** report potential sources of danger and wear appropriate safety gear to avoid accidents.
- GS10.** planning of work and using tools in an organized manner to avoid accidents/incidents.
- GS11.** schedule daily activities and drawing up priorities related to workplace safety.
- GS12.** use of safety equipment such as fire extinguisher during fire accidents
- GS13.** use personal protective equipments (ppes) such as protective gear, goggles, caps, shoes,etc.
- GS14.** follow code of safety conduct
- GS15.** manage relationships with co-workers and managers of the who may be stressed, frustrated,confused or angry
- GS16.** recognize problems and search for solutions
- GS17.** choose best methods to complete assigned tasks
- GS18.** identify problems immediately and take up solutions quickly to resolve delays.
- GS19.** assess situation & identify appropriate control measures.
- GS20.** critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities.
- GS21.** initiate and take-up own working and learning.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain Personal Health & Safety in biomass pellet manufacturing facility</i>	25	25	-	-
PC1. identify the requirements for safe work area	3	-	-	-
PC2. explain how to Administer first aid and demonstrate how to Administer first aid.	5	5	-	-
PC3. . identify the personal protective equipment used for the specific purpose	4	-	-	-
PC4. • identify the heavy electrical and mechanical equipment's associated with manufacturing and • show how to Identify the heavy electrical and mechanical equipment's associated with • bioenergy production manufacturing	5	5	-	-
PC5. • identify work safety procedures and instructions to operate bioenergy production machines • and demonstrate how to Identify and perform work safety procedures and instructions to • operate bioenergy production machines	5	5	-	-
PC6. • explain occupational health & safety standards and regulations for bioenergy production • manufacturing process	3	-	-	-
PC7. demonstrate good housekeeping practices and infection control guidelines	-	5	-	-
PC8. • demonstrate how to Dispose- off any waste materials in accordance with safe working • practices and procedures	-	5	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4051
NOS Name	Maintain Personal Health & Safety in biomass pellet manufacturing facility
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Process Line Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	22/01/2026
Next Review Date	22/01/2029
NSQC Clearance Date	22/01/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4013.Basics of Biomass availability and utilization	40	10	0	0	50	12
SGJ/N4014.Preparation of Raw material and quality assessment	23	27	0	0	50	12
SGJ/N4015.Equipment and machineries of Biomass pellet plant	30	20	0	0	50	12
SGJ/N4016.Manufacturing of Biomass pellet	30	20	0	0	50	12
SGJ/N4017.Operation and maintenance of various components of biomass pellet plant	30	20	0	0	50	13
SGJ/N4018.Package and Storage of biomass pellet	30	20	0	0	50	13
SGJ/N4051.Maintain Personal Health & Safety in biomass pellet manufacturing facility	25	25	0	0	50	13
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	13
Total	228	172	-	-	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.